

CS 5542

Controlled **Image Generation** & Speech Intelligence

Presented by: **Tina Nguyen**

QUIZ 1

Controlled E-Commerce Image Generation

CONTEXT & METHODOLOGY

The Challenge

E-commerce product photography is **prohibitively expensive**. While generative AI (SDXL) offers a solution, it is notoriously unpredictable in styling and structural consistency.

Our Methodology

We injected **metadata** (category, material, style) into a rigid prompt template, comparing *Naive Prompting* against *Structured Prompt Engineering*.

The Strategic Goal

To enforce **strict visual consistency** (lighting, background, and perspective) across a diverse and high-volume product catalog using deterministic evaluation.

QUANTITATIVE EVALUATION

Metric Type	Naive Prompting	Structured Prompting	Delta Improvement
Text Alignment (CLIP)	0.281	0.284	+1.1%
Visual Consistency (SSIM)	0.360	0.442	+22.8%
Structural Stability	Variable	Rigid/Consistent	High Impact

**Dataset: 10 diverse e-commerce products evaluated across 3 deterministic seeds.*

| CONSISTENCY IMPACT

+22.8%

Improvement in SSIM Scores

Structured prompts proved **exceptionally effective** for solid objects (mugs, sneakers) by stabilizing geometry and lighting.

Note: Complex geometries like shoes remain a challenge for current prompting techniques.

Baseline vs Structured prompt - Generated Samples
Product 1 - Baseline



Product 1 - Improved



Product 2 - Baseline



Product 2 - Improved



Product 3 - Baseline



Product 3 - Improved



QUIZ 2

AI Speech Intelligence Autonomous Pipeline

THE MULTIMODAL ARCHITECTURE



7-Model Chain

Orchestrated 7 different Hugging Face models into a unified processing loop.



Intelligence Loop

Full autonomous loop:
Speech to Text to Summary
to synthesized Voice.



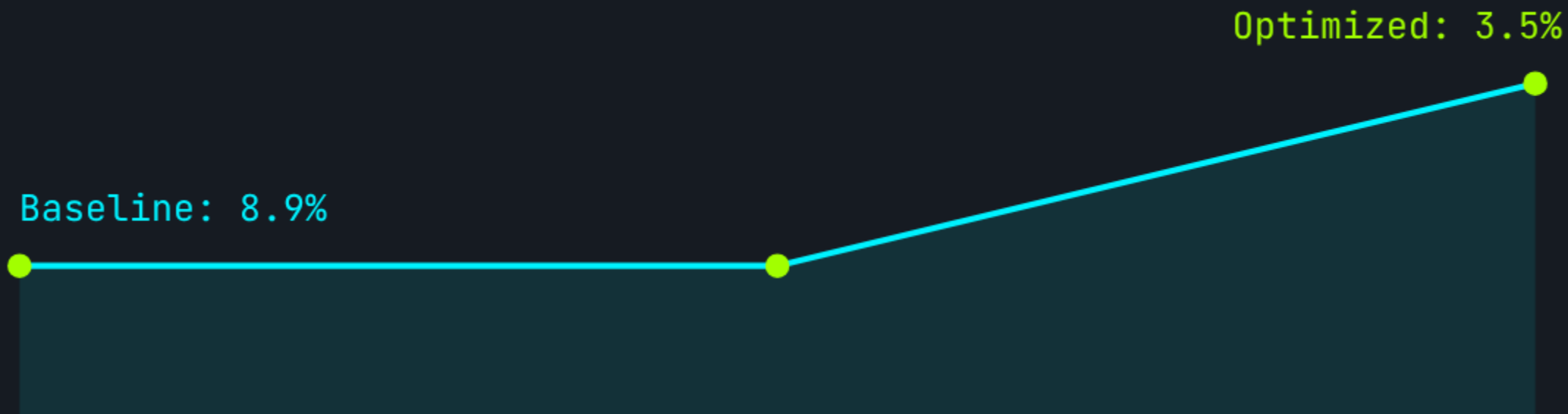
Structured Data

Outputting clean JSON with
regex-parsed action items
and sentiment analysis.

THE EXECUTION PIPELINE



OPTIMIZING WORD ERROR RATE (WER)



Adding a **domain-aware initial prompt** to Whisper reduced transcription errors by a massive **61%**.

Strategic prompting achieved this reduction **without retraining** the model or increasing compute latency.

EXTRA CREDIT: MULTIMODAL LOOP

The pipeline successfully closed the **multimodal loop** by synthesizing insights back into high-quality speech.

- ✓ Actionable JSON output
- ✓ Regex-parsed action items
- ✓ Spanish Translation synthesis



STRATEGIC TAKEAWAYS

- ✓ **Strategic Prompting is Paramount:** Whether in Vision (SDXL) or Audio (Whisper), structured prompts are the primary driver of reliability.
- ✓ **The Power of Orchestration:** Chaining modular components enables end-to-end autonomous processing for complex human-centric tasks.
- ✓ **Data-Driven Reproducibility:** Success was validated through deterministic seeds and rigorous metrics (SSIM, CLIP, and WER).

Thank You!

Any Questions?

Tina Nguyen